

Assignment No 01

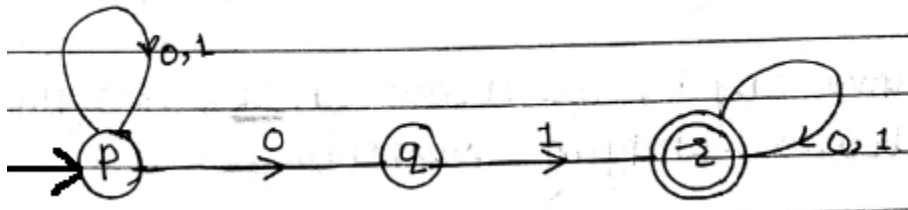
Module1: Basic Concepts and Finite Automata

Module2: Regular Expressions and Languages

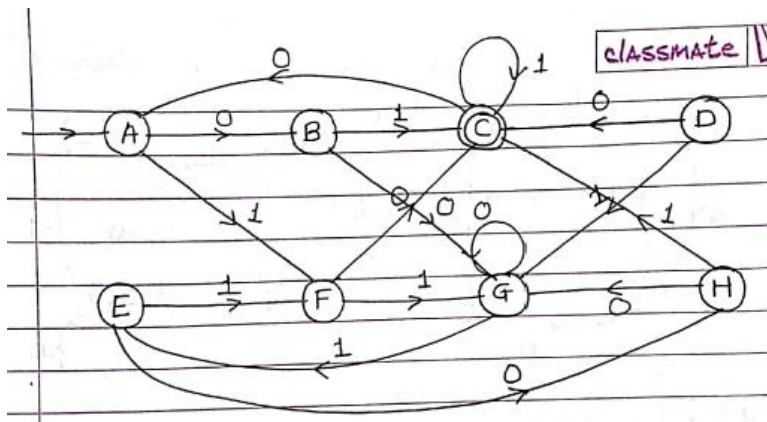
CO1	Understand concepts of Theoretical Computer Science, difference and equivalence of DFA and NFA, languages described by finite automata and regular expressions.
CO2	Understand the basics of Regular expressions and Regular Languages, and illustrate the equivalence of Finite automata and regular expressions.

1. Write short note on
 - a. DFA
 - b. NFA
 - c. Moore Machine
 - d. Mealy Machine

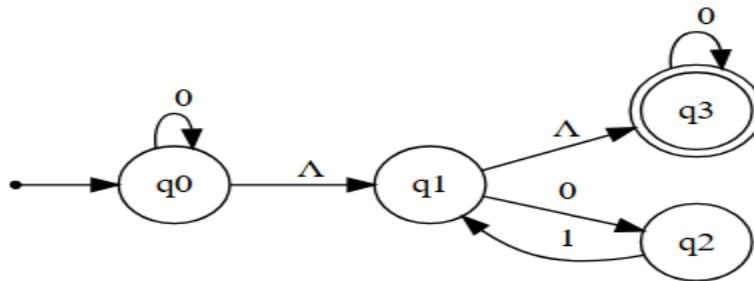
2. Convert NFA to DFA



3. Minimize the states of DFA



4. Convert following Epsilon NFA to Equivalent NFA



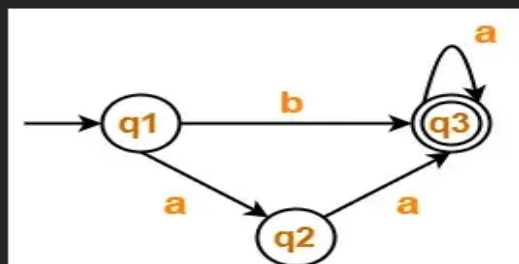
5. Write a short note on
- Arden's theorem
 - Pumping lemma for RL
 - Closures properties of RL
 - Decision Properties of Regular Languages
6. Define Following
- Regular Expression
 - Regular Language

7.

Give formal Definition of Pumping Lemma for Regular Language. Prove that the following language is not regular. $L = \{wrw^r \mid w \in \{a,b\}^*, r \in \{c\}, |w| \geq 1\}$

8.

Find regular expression for the following DFA using Arden's Theorem-



9.

Convert following RE to NFA-ε and convert it to minimised DFA corresponding to it
 $(0+11)^*(10)(11+0)^*$ [10]

Signature of Subject Teacher